

Information and Entertainment System

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware or equivalent diagnostic scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent
 ST3104-A	Back-probe Pins POM6411 or equivalent

DTC Charts

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

To retrieve DTCs , refer to Audio Control Module (ACM) Self-Diagnostic Mode.

Audio Front Control Module (ACM) DTC Chart

DTC	Description	Action
B1342	Internal Failure: Flash ROM Checksum Error	NOTE: Repair all other DTCs first before diagnosing DTC B1342. CLEAR the DTCs. REPEAT the self-test. If only DTC B1342 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM).
B2965	Speaker Failure	- For a LL door speaker concern, GO to Pinpoint Test D. - For a RF door speaker concern, GO to Pinpoint Test E. - For a RR door speaker concern (if equipped), GO to Pinpoint Test F.

DTC	Description	Action
		For a LR door speaker concern (if equipped), GO to Pinpoint Test G.
B2924	Stuck Button	GO to Pinpoint Test A .

Symptom Charts

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Symptom Chart — General Audio System

Condition	Possible Causes	Action
One or more ACM buttons are inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test A.
The ACM illumination is inoperative or does not dim	- Illumination system concern ACM	REFER to Section 413-00.

Symptom Chart — Sound Quality

Condition	Possible Causes	Action
Poor reception — AM/FM	Refer to the Pinpoint Test	GO to Pinpoint Test B.
Continuous seek or scan — AM/FM	Refer to the Pinpoint Test	GO to Pinpoint Test B.
No sound from all speakers	Refer to the Pinpoint Test	GO to Pinpoint Test C.
The LF door speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test D.
The RF door speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test E.
The RR door speaker is inoperative (if equipped)	Refer to the Pinpoint Test	GO to Pinpoint Test F.
The LR door speaker is inoperative (if equipped)	Refer to the Pinpoint Test	GO to Pinpoint Test G.
Poor Sound Quality Or Distorted Sound From One Or More Speakers (Not All Speakers)	Refer to the Pinpoint Test	GO to Pinpoint Test H.

Pinpoint Tests

Pinpoint Test A: One or more ~~ACM~~ Buttons Are Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ACM in Information and Entertainment System.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B2924	Stuck Button	Set by the ACM when buttons are held for more than 100 milliseconds during the self-test.

Possible Sources

- Debris
- ACM

PINPOINT TEST A : ONE OR MORE ACM BUTTONS ARE INOPERATIVE

A1 CHECK BUTTON OPERATION	
• Operate all of the ACM buttons and check for proper operation.	
Do all buttons work properly?	
Yes	The system is operating correctly at this time.
No	GO to A2 .
A2 CHECK FOR DEBRIS	
• CHECK the ACM buttons for dust collection and/or debris.	
Was any dust collection and/or debris found?	
Yes	Clean as necessary
No	GO to A3 .
A3 RECHECK FOR DTC B2924	
• Ignition ON. • Clear the ACM DTCs. REFER to Audio Control Module (ACM) Self-Diagnostic Mode. • Ignition OFF.	

- Ignition ON.
- Wait at least 2 minutes.
- Retrieve all DTCs.

Is DTC B2924 set?

Yes	GO to A4 .
No	The system is operating correctly at this time.

A4 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test B: Poor Reception Or Continuous Seek Or Scan — AM/FM

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to AM/FM Radio, **ACM** , and Antenna in Information and Entertainment System.

Possible Sources

- AM/FM antenna
- AM/FM antenna coaxial cable

- Charging system
- Ignition system
- Noise suppression equipment
- Wiring, terminals, or connectors
- ACM

PINPOINT TEST B : POOR RECEPTION OR CONTINUOUS SEEK OR SCAN — AM/FM

B1 CHECK THE AUDIO SYSTEM RECEPTION

- Operate the audio system in radio tuner AM/FM mode.
- Check the reception with the engine running, and with the engine off.

Does the poor reception only occur with the engine running?

Yes	GO to B2 .
No	GO to B4 .

B2 CHECK THE GENERATOR

- Ignition OFF.
- Disconnect: Wiring harness to the generator voltage regulator.
- Start the engine.
- Operate the audio system in radio tuner AM/FM mode.

Is the reception OK?

Yes	INSTALL a new generator. REFER to Section 414-00.
No	GO to B3 .

B3 CHECK THE IGNITION CIRCUITS

- Ignition OFF.
- Connect: Wiring harness to the generator voltage regulator.
- Visually inspect the engine compartment and make sure all ignition coils are correctly and securely connected, and that there are no visible cracks in the coil housings.
- Inspect all the wiring harnesses and connectors for damaged insulation and loose or broken conditions.

Are the ignition components OK?

Yes	USE a jumper cable to ground various parts of the vehicle to the frame (for example: engine, fenders, quarter panels, stone deflectors, air cleaner, and body sheet metal). When the noise is eliminated, PROVIDE a permanent ground where necessary.
No	REPAIR the ignition system as necessary.

B4 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: AM/FM antenna coaxial cable at the **ACM** . Refer to Audio Control Module (ACM).
- Disconnect: AM/FM antenna inline coaxial cable connection behind the glove compartment. Refer to Antenna Cable — AM/FM.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground
AM/FM antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.
No	GO to B5 .

B5 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to B6 .
No	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

B6 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM coaxial cable core at the ACM	—	Lower AM/FM antenna coaxial cable core at the inline connection	—
AM/FM antenna coaxial cable shield at the ACM	—	Lower AM/FM antenna coaxial cable shield at the inline connection	—

Is the resistance less than 1 ohm?

Yes	GO to B7 .
No	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

B7 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to B8 .
No	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

B8 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: AM/FM antenna inline coaxial cable connection behind the glove compartment.
- Disconnect: AM/FM antenna coaxial cable at the antenna unit. Refer to Antenna — AM/FM.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead	Negative Lead
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Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground
AM/FM antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.
No	GO to B9 .

B9 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to B10 .
No	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

B10 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable core at the antenna unit (harness side)	—
AM/FM antenna coaxial cable shield at the ACM	—	AM/FM antenna coaxial cable shield at the antenna unit (harness side)	—

Is the resistance less than 1 ohm?

Yes	GO to B11 .
No	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

B11 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to B12 .
No	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

B12 CHECK THE AM/FM ANTENNA UNIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: AM/FM antenna coaxial cable at the antenna unit.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground
AM/FM antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.
No	GO to B13 .

B13 CHECK THE AM/FM ANTENNA UNIT CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to B14 .
No	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.

B14 CHECK THE AM/FM ANTENNA UNIT CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to B15 .
No	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.

B15 CHECK THE AM/FM ANTENNA UNIT SHIELD FOR CONTINUITY TO GROUND

- Disconnect: AM/FM antenna coaxial cable at the antenna unit.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna unit shield at the antenna unit (component side)	—	—	Ground

Is the resistance less than 1 ohm?

Yes	GO to B17 .
No	GO to B16 .

B16 CHECK THE AM/FM ANTENNA UNIT SHIELD FOR CONTINUITY TO ANTENNA UNIT BASE

- Remove the AM/FM antenna. Refer to Antenna — AM/FM.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna unit shield at the antenna unit (component side)	—	AM/FM antenna base (unpainted surface)	—

Is the resistance less than 1 ohm?

Yes	GO to B17 .
No	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.

B17 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ~~ACM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ~~ACM~~ connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ~~ACM~~ in Information and Entertainment System.

When the ignition is in the START position, the ~~ACM~~ receives voltage on the START signal circuit and mutes the speaker output. This prevents voltage spikes from producing popping noises through the speakers. A short to voltage on this circuit causes the ~~ACM~~ output to be muted continuously.

Possible Sources

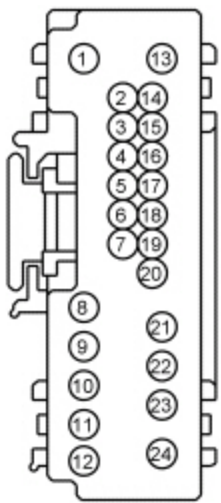
- Wiring, terminals or connectors
- ~~ACM~~

PINPOINT TEST C : NO SOUND FROM ALL SPEAKERS

C1 CHECK THE START SENSE CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ~~ACM~~ [C290A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C290A Pin 15	CDC54 (WH/GN)	—	Ground



Is the voltage **5V** or greater?

Yes	REPAIR the circuit. TEST the system for normal operation.
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No	GO to C2 .
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C2 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test D: The LF Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to **ACM** in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B2965	Speaker Failure	Sets when a short to ground, short to voltage, or short together is detected on the LF , RF , RR (if equipped), or LR (if equipped) door speaker circuits.

Possible Sources

- Wiring, terminals, or connectors

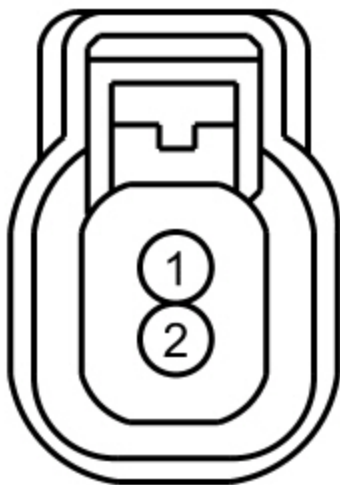
- Speaker
- ACM

PINPOINT TEST D : THE LF DOOR SPEAKER IS INOPERATIVE

D1 CHECK THE AUDIO SIGNAL TO THE LF DOOR SPEAKER

- Ignition OFF.
- Disconnect: LF Door Speaker [C523](#).
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	C523 Pin 2	RME07 (WH/BN)



Is a fluctuating AC voltage present?

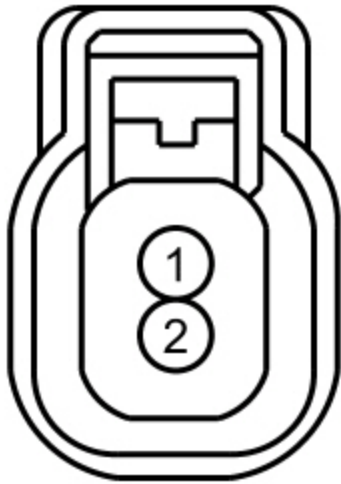
Yes	INSTALL a new LF door speaker. REFER to Speaker — Door.
No	GO to D2 .

D2 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C290A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit

C523 Pin 1	VME07 (WH)	—	Ground
C523 Pin 2	RME07 (WH/BN)	—	Ground



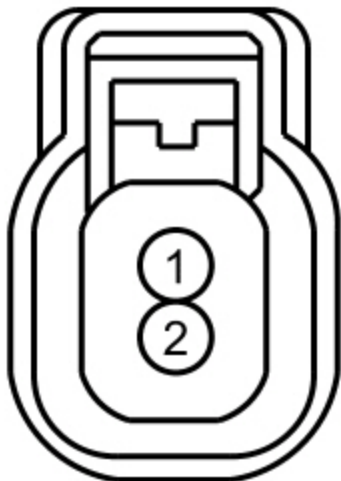
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to D3 .

D3 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	—	Ground
C523 Pin 2	RME07 (WH/BN)	—	Ground



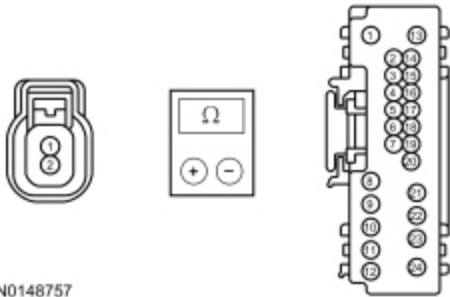
Are the resistances greater than 10,000 ohms?

Yes	GO to D4 .
No	REPAIR the circuit in question.

D4 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	C290A Pin 8	VME07 (WH)
C523 Pin 2	RME07 (WH/BN)	C290A Pin 21	RME07 (WH/BN)



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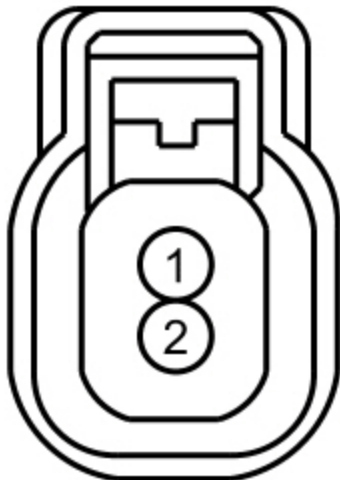
Are the resistances less than 3 ohms?

Yes	GO to D5 .
No	REPAIR the circuit in question.

D5 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	C523 Pin 2	RME07 (WH/BN)



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Are the resistances greater than 10,000 ohms?

Yes	GO to D6 .
No	REPAIR the circuit in question.

D6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test E: The ~~RF~~ Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ACM in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B2965	Speaker Failure	Sets when a short to ground, short to voltage, or short together is detected on the LF , RF , RR (if equipped), or LR (if equipped) door speaker circuits.

Possible Sources

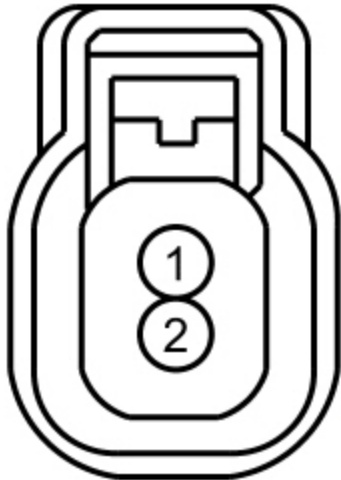
- Wiring, terminals, or connectors
- Speaker
- ~~ACM~~

PINPOINT TEST E : THE RF DOOR SPEAKER IS INOPERATIVE

E1 CHECK THE AUDIO SIGNAL TO THE RF DOOR SPEAKER

- Ignition OFF.
- Disconnect: ~~RF~~ Door Speaker [C612](#).
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	C612 Pin 2	RME10 (WH/OG)



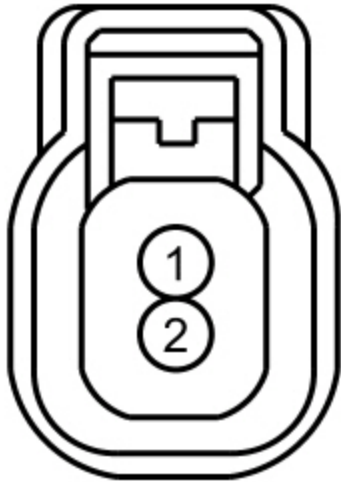
Is a fluctuating AC voltage present?

Yes	INSTALL a new RF door speaker. REFER to Speaker — Door.
No	GO to E2 .

E2 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C290A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	—	Ground
C612 Pin 2	RME10 (WH/OG)	—	Ground



Is any voltage present?

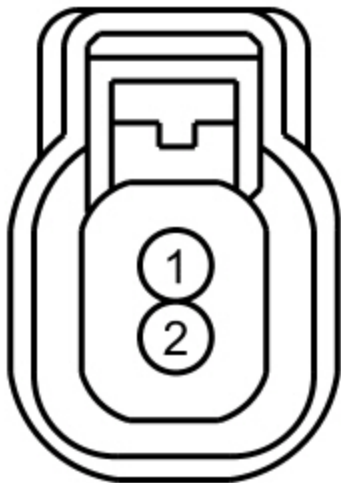
Yes	REPAIR the circuit in question.
No	GO to E3 .

E3 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
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Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	—	Ground
C612 Pin 2	RME10 (WH/OG)	—	Ground



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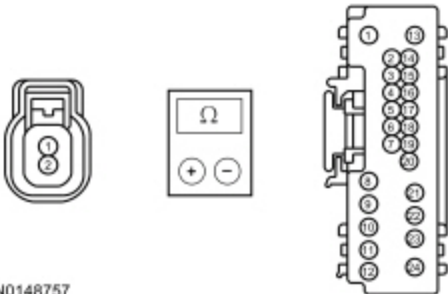
Are the resistances greater than 10,000 ohms?

Yes	GO to E4 .
No	REPAIR the circuit in question.

E4 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	C290A Pin 11	VME10 (WH/VT)
C612 Pin 2	RME10 (WH/OG)	C290A Pin 12	RME10 (WH/OG)



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Are the resistances less than 3 ohms?

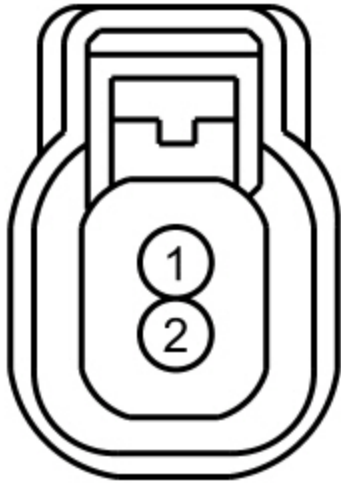
Yes	GO to E5 .
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No	REPAIR the circuit in question.
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E5 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	C612 Pin 2	RME10 (WH/OG)



Are the resistances greater than 10,000 ohms?

Yes	GO to E6 .
No	REPAIR the circuit in question.

E6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ~~ACM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ~~ACM~~ connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM .
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	REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test F: The RR Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ACM in Information and Entertainment System.

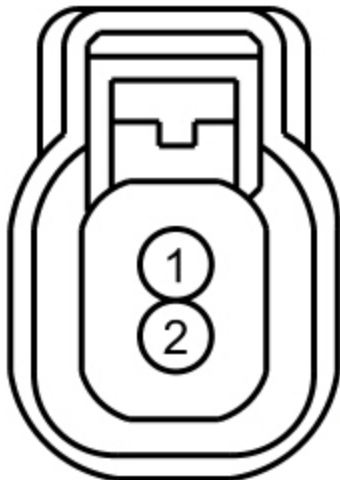
Root DTC Description	Failure Types	Fault Trigger Conditions
B2965	Speaker Failure	Sets when a short to ground, short to voltage, or short together is detected on the LF , RF , RR (if equipped), or LR (if equipped) door speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- ACM

PINPOINT TEST F : THE RR DOOR SPEAKER IS INOPERATIVE

F1 CHECK THE AUDIO SIGNAL TO THE RR DOOR SPEAKER			
• Ignition OFF. • Disconnect: RR Door Speaker C802. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:			
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	C802 Pin 2	RME12 (BN/BU)



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Is a fluctuating AC voltage present?

Yes	INSTALL a new RR door speaker. REFER to Speaker — Door.
No	GO to F2 .

F2 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C290A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	—	Ground
C802 Pin 2	RME12 (BN/BU)	—	Ground



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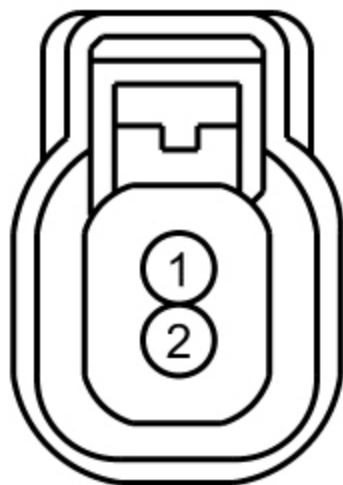
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to F3 .

F3 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	—	Ground
C802 Pin 2	RME12 (BN/BU)	—	Ground



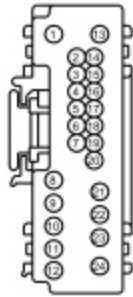
Are the resistances greater than 10,000 ohms?

Yes	GO to F4 .
No	REPAIR the circuit in question.

F4 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	C290A Pin 10	VME12 (BN/WH)
C802 Pin 2	RME12 (BN/BU)	C290A Pin 23	RME12 (BN/BU)



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Are the resistances less than 3 ohms?

Yes	GO to F5 .
No	REPAIR the circuit in question.

F5 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	C802 Pin 2	RME12 (BN/BU)



Are the resistances greater than 10,000 ohms?

Yes	GO to F6 .
No	REPAIR the circuit in question.

F6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test G: The **LR Door Speaker Is Inoperative**

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to **ACM** in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B2965	Speaker Failure	Sets when a short to ground, short to voltage, or short together is detected on the LF , RF , RR (if equipped), or LR (if equipped) door speaker circuits.

Possible Sources

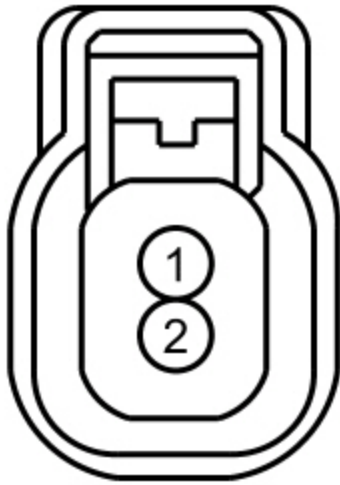
- Wiring, terminals, or connectors
- Speaker
- **ACM**

PINPOINT TEST G : THE LR DOOR SPEAKER IS INOPERATIVE

G1 CHECK THE AUDIO SIGNAL TO THE LR DOOR SPEAKER

- Ignition OFF.
- Disconnect: LR Door Speaker [C702](#).
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	C702 Pin 2	RME09 (BN/YE)



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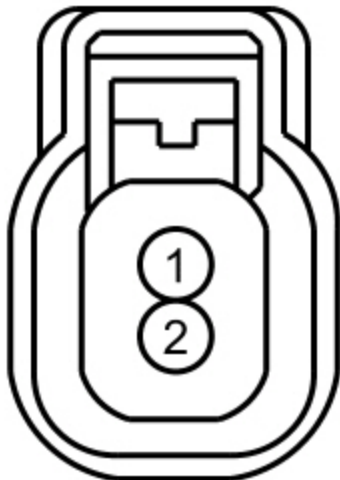
Is a fluctuating AC voltage present?

Yes	INSTALL a new LR door speaker. REFER to Speaker — Door.
No	GO to G2 .

G2 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C290A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	—	Ground
C702 Pin 2	RME09 (BN/YE)	—	Ground



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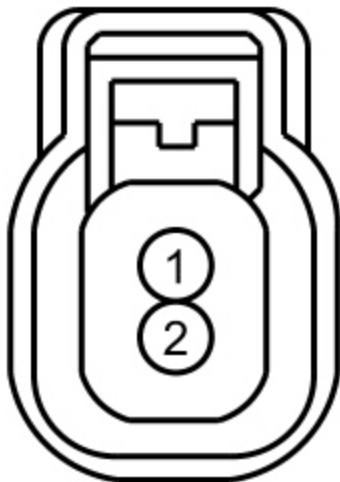
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to G3 .

G3 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	—	Ground
C702 Pin 2	RME09 (BN/YE)	—	Ground



•

Are the resistances greater than 10,000 ohms?

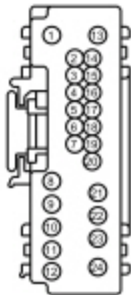
Yes	GO to G4 .
-----	----------------------------

No	REPAIR the circuit in question.
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G4 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	C290A Pin 9	VME09 (WH/GN)
C702 Pin 2	RME09 (BN/YE)	C290A Pin 22	RME09 (BN/YE)



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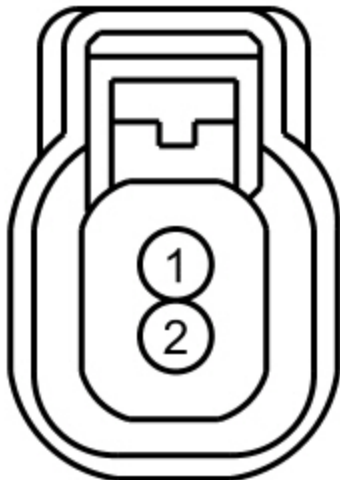
Are the resistances less than 3 ohms?

Yes	GO to G5 .
No	REPAIR the circuit in question.

G5 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	C702 Pin 2	RME09 (BN/YE)



•

Are the resistances greater than 10,000 ohms?

Yes	GO to G6 .
No	REPAIR the circuit in question.

G6 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test H: Poor Sound Quality Or Distorted Sound From One Or More Speakers (Not All Speakers)

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation

The ACM sends audio signals to the speakers in the form of AC voltage, resulting in clear audio output.

Possible Sources

- Loose component fasteners
- Loose door handle, lock, and other trim panel components and fasteners
- Loose items in storage areas
- Loose speaker grilles
- Loose wire harnesses or wire harness fasteners
- Speaker(s)
- Water shield bonding and placement

Visual Inspection and Diagnostic Pre-checks

Visually inspect around the suspect area for any possible rattle conditions such as:

- Any storage compartments including cup holders (if equipped)
- Door map pockets for contents that can rattle (if equipped)
- Door pull cup attachments (if equipped)
- Door switch bezels (if equipped)
- Package tray (if equipped)
- Speaker grilles
- Trim panels and fasteners

PINPOINT TEST H : POOR SOUND QUALITY OR DISTORTED SOUND FROM ONE OR MORE SPEAKERS (NOT ALL SPEAKERS)

H1 ISOLATE THE ZONE	
• Operate the audio system using digital media (CD, MP3, etc.). • Using the audio system fade and balance feature, adjust the audio sound to each of the four zones (LF , RF , LR , and RR) of the vehicle to isolate the poor sound quality. • Locate and apply pressure to the trim panel(s) around the poor sound quality area in question.	
Does applying pressure to a trim panel reduce or eliminate the audible noise?	
Yes	REPAIR or REPLACE the trim panel as needed.
No	GO to H2 .

H2 REMOVE AND INSPECT BEHIND/UNDERNEATH THE SUSPECT TRIM PANEL(S)	
• Remove the trim panel to access the suspect speaker. • Operate the audio system using digital media (CD, MP3, etc.). • Validate where the poor sound quality area is located. • Check: ▪ Trim panel for loose components around the speaker area such as the speaker grille ▪ Trim panel joining components for missing or broken pieces ▪ Lock and handle mechanical parts for correct attachment	

- Wire harnesses for correct routing
- Wire harness fasteners for correct attachment
- Water shield for correct placement (speaker air path not blocked)
- Water shield for correct bonding to sheet metal
- Storage areas for loose items
- All child safety belt anchors (if equipped)
- All safety belt retractors
- All speaker bracket fasteners and other fasteners are secured and tightened to specified torque

Was the source of the noise located?

Yes	REPAIR or REPLACE any loose or broken component or fastener as needed.
No	GO to H3 .

H3 CHECK THE SUSPECT SPEAKER FOR WATER INTRUSION

- Inspect for watermarks.
- Check the:
 - Cone
 - Magnet
 - Basket

Are any watermarks present on the speaker?

Yes	VERIFY the water shield is in the correct location, REPAIR or REPLACE any trim, door, or speaker seal as required. DRY the speaker in question and TEST the system for normal operation.
No	GO to H4 .

H4 ISOLATE SPEAKER TO VERIFY NOISE

- Remove the suspect speaker from its location and leave the speaker connected to the harness.
- Hold the speaker away from any trim panel(s) and ensure the speaker is isolated from contact.
- Operate the audio system using digital media (CD, MP3, etc.).

Is the noise still present in the suspect speaker?

Yes	INSTALL a new speaker for the one in question.
No	LOCATE the source of the noise and REPAIR as needed.